

BALAGURU K

SOFTWARE DEVELOPER

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SKILLS

- Python, Java Programming
- Data Structures & Algorithms
- Object Oriented Programming
- DBMS & SQL Programming
- HTML, CSS, JavaScript
- Flask API
- Git and GitHub
- AWS Cloud Fundamentals
- Agile Methodology
- Debugging & Unit Testing (Basics)
- Machine and Deep Learning with Python

CERTIFICATIONS

- Infosys Springboard: Python , Java
- SkillRack : Python & C Programmer
- Infosys Springboard : React.js, Springboot
- Udemy : AWS Fundamentals
- Infosys Springboard : ML & DL
- NPTEL : IOT(80%)

INTERSHIPS

MSME | Jan 2025 - Mar 2025

Developed a Python-based web application using HTML, CSS, and JavaScript for the frontend and Flask for the backend. Implemented clean coding practices, ensuring maintainability, reliability, and efficient end-to-end system functionality. Designed and integrated frontend and backend components to deliver a seamless user experience.

Infosys Springboard | Nov 2025 - Dec 2025

Developed a Forest Fire Weather Index (FWI) prediction system with a React.js frontend and Flask backend, integrating an ML model trained on real-world datasets. Built data preprocessing pipelines and delivered a fully functional application supporting environmental risk detection, following Agile practices throughout.



PROFILE

Software Engineering graduate with strong foundations in Python, Java, SQL, and web development using Flask, HTML, CSS, and JavaScript. Experienced in developing web applications, AI-based solutions, and data-driven projects through internships, academic work, and hackathons. Familiar with AWS Cloud, Git, and Agile development practices, with a keen interest in building scalable and efficient software solutions. Committed to writing clean, maintainable code, solving real-world problems, and continuously learning emerging technologies to deliver high-quality software products.



PROJECTS

• AI-Powered Agricultural Chatbot for Crop Health Monitoring

Technologies Used : Python, Flask, MySQL, HTML, CSS, JavaScript
Built a full-stack web application featuring an AI-driven chatbot that processes user inputs and crop images to deliver real-time disease insights and crop recommendations. Designed and implemented REST APIs, managed database operations with MySQL, and developed image processing workflows ensuring clean backend architecture, reliable data flow, and maintainable code throughout.

• AWS Serverless Architecture Implementation

Technologies Used : HTML, CSS , JavaScript , AWS Amplify, API Gateway , Lambda , DynamoDB
Designed and deployed a serverless web application using AWS Lambda and API Gateway for modular, scalable backend logic. Implemented secure access control, efficient data storage with DynamoDB, and clean API design to ensure reliable communication between frontend and cloud backend components.

• AI-Powered PDF Document Question Answering System

Technologies Used : Python, AWS S3, Titan Embedding, FAISS , LLM · Streamlit
Built an end-to-end document intelligence pipeline that splits and processes PDFs, generates vector embeddings using AWS Titan, and stores them in FAISS for fast semantic search. Integrated an LLM-based query engine to retrieve context-relevant content and deliver real-time, accurate answers from large documents.



EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Business Systems 2022-2026

S.A. Engineering College, Chennai | CGPA : 8.31 / 10

Higher Secondary Certificate (HSC) – 12th Standard

2021-2022

Krishnaswamy Matric Hr. Sec. School, Cuddalore

Group: Computer Science | Marks : 493 / 600 (82.17%)

LANGUAGES

- Tamil
- English